3.2.P.5.3 Validation of Analytical Procedures [2439-exNA, Intrathecal Injection, Argonaut]

The analytical methods used for the release of 2439-exNA drug product are compendial and in compliance with the USP (United States Pharmacopeia) standards, except assay by UV/Visible spectrophotometry, identification, purity and impurities analysis by both denaturing AEX (anion exchange chromatography) and non-denaturing SEC (size-exclusion chromatography). The non-denaturing and denaturing methods were developed by the drug substance manufacturer (Section 3.2.S.4.2) and then were transferred to the drug product manufacturer. The extinction coefficient was determined in order to perform the UV assay. Method validation was performed for the denaturing and non-denaturing analytical methods. The validation information for these methods is presented in this section.

A summary of the method validations performed for the novel chromatographic methods is presented in Table 1.

Table 1. Validation of Analytical Procedures Non-Denaturing and Denaturing Analytical Methods

Test	Method	Validation Parameters Assessed	Location
Identity, Purity, Impurities	Non-Denaturing SEC chromatography	Comparability of retention time and peak areas for drug product with the precursor drug substance, Limit of Quantification (LOQ)	Table 2
Purity, Impurities	Denaturing AEX chromatography	Comparability of retention time and peak areas for drug product with the precursor drug substance, Limit of Quantification (LOQ)	Table 3

SEC = Size Exclusion Chromatography; AEX = Anion Exchange Chromatography

3.2.P.5.3.1 Non-denaturing SEC Identity, Purity and Impurities

The non-denaturing SEC method for identity, purity and impurity analysis of 2439-exNA drug product was used for testing the drug product at the drug substance manufacturer. The method was validated for drug product samples. The validation confirms the suitability of the validated drug substance method for use in testing drug product. The results of the validation are summarized in Table 2. All experiments met the pre-defined method validation protocol acceptance criteria.

Table 2. Summary of Non-denaturing SEC Method Results for 2439-exNA drug product

Method Property	Acceptance Criteria	Results	Pass/Fail (or report)
Absolute difference in retention time between drug product peak and precursor drug substance	Absolute difference ≤0.2 min	[redacted]	Pass
Ratio of purity (area%) between drug substance and drug product	98-102%	[redacted]	Pass
Ratio of impurities by RRT between drug substance and drug product	report	[redacted]	N/A
Quantitation Limit (LOQ) at 0.2% nominal concentration	The S/N of the main peak in all LOQ solutions ≥10	[redacted]	Pass
	The RSD% of antisense and sense peak areas in all LOQ solutions ≤50%	[redacted]	Pass

S/N = signal to noise ratio; RSD = relative standard deviation; RRT = relative retention time

3.2.P.5.3.1.1 Representative chromatograms

Figure 1. Blank (Diluent [water] Injection)

Figure 2. Representative Chromatogram for 2439-exNA drug product

3.2.P.5.3.2 Denaturing AEX Identity, Purity and Impurities

The denaturing AEX method for identity, purity and impurity analysis of 2439-exNA drug product has been transferred from the drug substance vendor. The transfer protocol demonstrates specificity, linearity, precision, and limit of quantitation (LOQ). The results of the method transfer are summarized in Table 3. All experiments met the pre-defined method transfer protocol acceptance criteria.

Table 3. Summary of Denaturing AEX Method results for 2439-exNA drug product

Method Property	Acceptance Criteria	Results	Pass/Fail (or report)
Absolute difference in retention time	Antisense and sense absolute difference ≤0.2 min	[redacted]	Pass
Ratio of purity (area%) between drug substance	98-102%	[redacted]	Pass

Method Property	Acceptance Criteria	Results	Pass/Fail (or report)
and drug product Ratio of impurities by RRT between drug substance and drug product	report	[redacted]	N/A
Quantitation Limit (LOQ)	The S/N of antisense and sense peaks in LOQ solutions ≥10	[redacted]	Pass
	The RSD% of antisense and sense peak areas in all LOQ solutions ≤50%	[redacted]	Pass

USP = United States pharmacopeia; NMT = not more than; S/N = signal to noise ratio; RSD = relative standard deviation; RT = retention time

3.2.P.5.3.1.1 Representative chromatograms

Figure 3. Blank (Diluent [water] Injection)

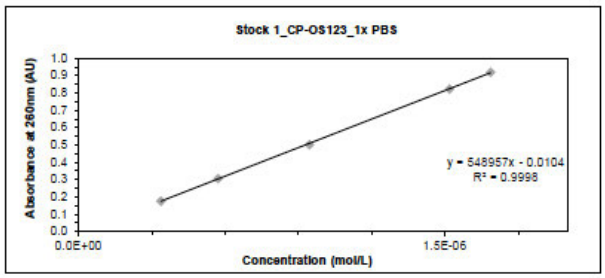
[redacted]

Figure 4. Representative Chromatogram for 2439-exNA drug product by denaturing AEX method

3.2.P.5.3.3 Assay by UV/Visible Spectrophotometry

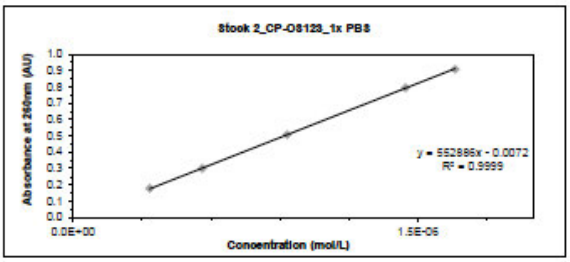
The extinction coefficient at 260 nm was determined for 2439-exNA in 1x PBS in triplicate at 5 concentration levels to be 282059.46 $M^{-1}cm^{-1}$ (individual determinations were 548957, 552886 and 545476 $M^{-1}cm^{-1}$ [RSD% 0.675%]). The conversion factor for use in the assay calculation is 22.01 OD/mg. Individual determinations had a R^2 of 0.9995, 0.9998 and 0.9999.

Figure 5. Determination of Extinction Coefficient



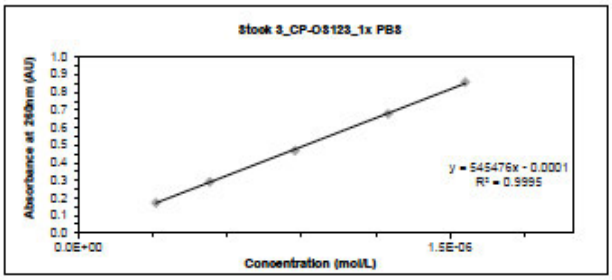
Result of Stock 1_CP-OS123_1x PBS

Sample ID	ε (L·mol−1·cm−1)
Stock 1_CP-OS123_1x PBS	548957



Result of Stock 2_CP-OS123_1x PBS

Sample ID	ε (L·mol−1·cm−1)
Stock 2_CP-OS123_1x PBS	552886



Result of Stock 3_CP-OS123_1x PBS

Sample ID	ε (L·mol−1·cm−1)
Stock 3_CP-OS123_1x PBS	545476